

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	28 September 2026
Team ID	SWTID-2026-2313
Project Name	AI BlogNest API
Maximum Marks	4 Marks

Technical Architecture

AI BLOGNEST API — TECHNOLOGY STACK

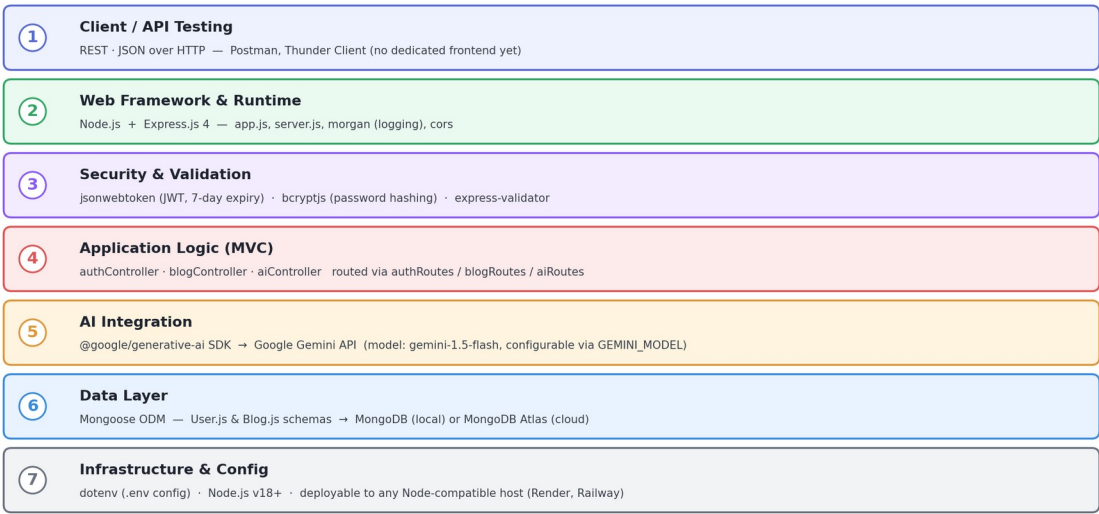


Figure 1: AI BlogNest API backend architecture diagram

Table-1: Components & Technologies

S.No.	Component	Description	Technology
1.	User Interface	How a client interacts with the application — currently API-only, tested via Postman / Thunder Client.	REST / JSON over HTTP
2.	Application Logic-1 (Auth)	Registration, login, and route protection.	jsonwebtoken, bcryptjs
3.	Application Logic-2 (Blog Management)	Blog CRUD logic and ownership enforcement.	Node.js, Express.js
4.	Application Logic-3 (AI Orchestration)	Builds prompts, calls Gemini, and cleans up generated text.	@google/generative-ai (Gemini)
5.	Database	Stores Users and Blogs collections.	MongoDB + Mongoose ODM

S.No.	Component	Description	Technology
6.	Cloud Database	Optional managed database hosting.	MongoDB Atlas
7.	File Storage	Not implemented in the current version (no file/image uploads).	N/A
8.	External API-1	AI-powered blog content generation and summarization.	Google Gemini API
9.	External API-2	Not implemented in the current version.	N/A
10.	Machine Learning Model	Large language model used for text generation, accessed via API (not self-hosted).	gemini-1.5-flash (default)
11.	Infrastructure (Server / Cloud)	Local Node.js server for development; deployable to any Node-compatible host. Local Server Configuration: Node.js v18+, 8 GB RAM. Cloud Server Configuration: Not yet deployed.	Local / Cloud (e.g. Render, Railway) + MongoDB Atlas

Table-2: Application Characteristics

S.No.	Characteristics	Description	Technology
1.	Open-Source Frameworks	Web framework and ODM used to build the API.	Express.js, Mongoose
2.	Security Implementations	Password hashing, token-based auth, input validation, and centralized error handling.	bcrypt, JWT, express-validator
3.	Scalable Architecture	Stateless 3-tier REST architecture (routes → controllers → models); Express instances can scale horizontally.	Node.js, Express.js
4.	Availability	Can run under a process manager for restart-on-crash; MongoDB Atlas offers	PM2 (optional), MongoDB Atlas

S.No.	Characteristics	Description	Technology
		managed uptime.	
5.	Performance	Non-blocking async I/O for all routes; JSON payloads kept small; latency is dominated by the external Gemini call, not the database.	Node.js event loop, MongoDB